



BITS Pilani
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Concrete Mix Design

Quality of Concrete depends upon



- Chemical composition
- Aggregate
- Water
- Admixtures
- Proportions
- Mixing
- Transporting
- Hydration
- Placing
- Vibrating
- Curing

Order of Operations for Concrete

✓ Specific operations must be performed in a certain order

✓ Final quality is influenced by every step

I. mix design (proportioning)

II. trial mixes & testing

III. batching

-----start the clock

IV. mixing

V. transporting

VI. pouring (placing)

VII. vibrating (consolidating)

Sampling and testing

-----initial set here

VIII. finishing

-----final set here

IX. curing

X. maintenance

Proportioning of Concrete Mixes (Mix Design)



- ❖ Mix design can be defined as the process of selecting suitable ingredients of concrete and determining their relative quantities with the purpose of producing and ***economical*** concrete which has certain minimum properties, notably ***workability, strength, and durability***
- ❖ Remains largely an empirical procedure
- ❖ Most design procedure are based on achieving a specified compressive strength at some given workability and age; it is assumed that if this is done, other properties will also be satisfactory



IS code method of Mix Design

- ✓ Step 1: Determining **target strength** for mix proportioning
- ✓ Step 2: Selection of **W/C ratio**
- ✓ Step 3: Selection of **water content**
- ✓ Step 4: Calculation of **cement content** based on W/C ratio
- ✓ Step 5: Determining **proportion of volume of coarse and fine aggregates**
- ✓ Step 6: **Mix calculations**

Please refer to IS 10262 for details

Strength Requirement



- Structural engineer “specifies” a strength of concrete used for design calculations – f_{ck}
- Concrete strength is variable
- Material engineer designs concrete so only a small proportion of the concrete will have a strength less than the strength assumed by the structural engineer.

Recall: Statistics

- Compressive strength test on 12 concrete samples: 23.8, 24.3, 20.7, 26.2, 24.1, 23.4, 26.8, 22.7, 19.4, 23.4, 21.5, and 19.5 MPa

$$\begin{aligned}\bar{X} &= \frac{\sum x}{n} \\ &= \frac{275.8}{12} \\ &= 22.98 \text{ MPa}\end{aligned}$$

$$\begin{aligned}s &= \sqrt{\frac{\sum (x - \bar{X})^2}{(n-1)}} \\ &= \sqrt{\frac{61.38}{11}} \\ &= 2.36 \text{ MPa}\end{aligned}$$

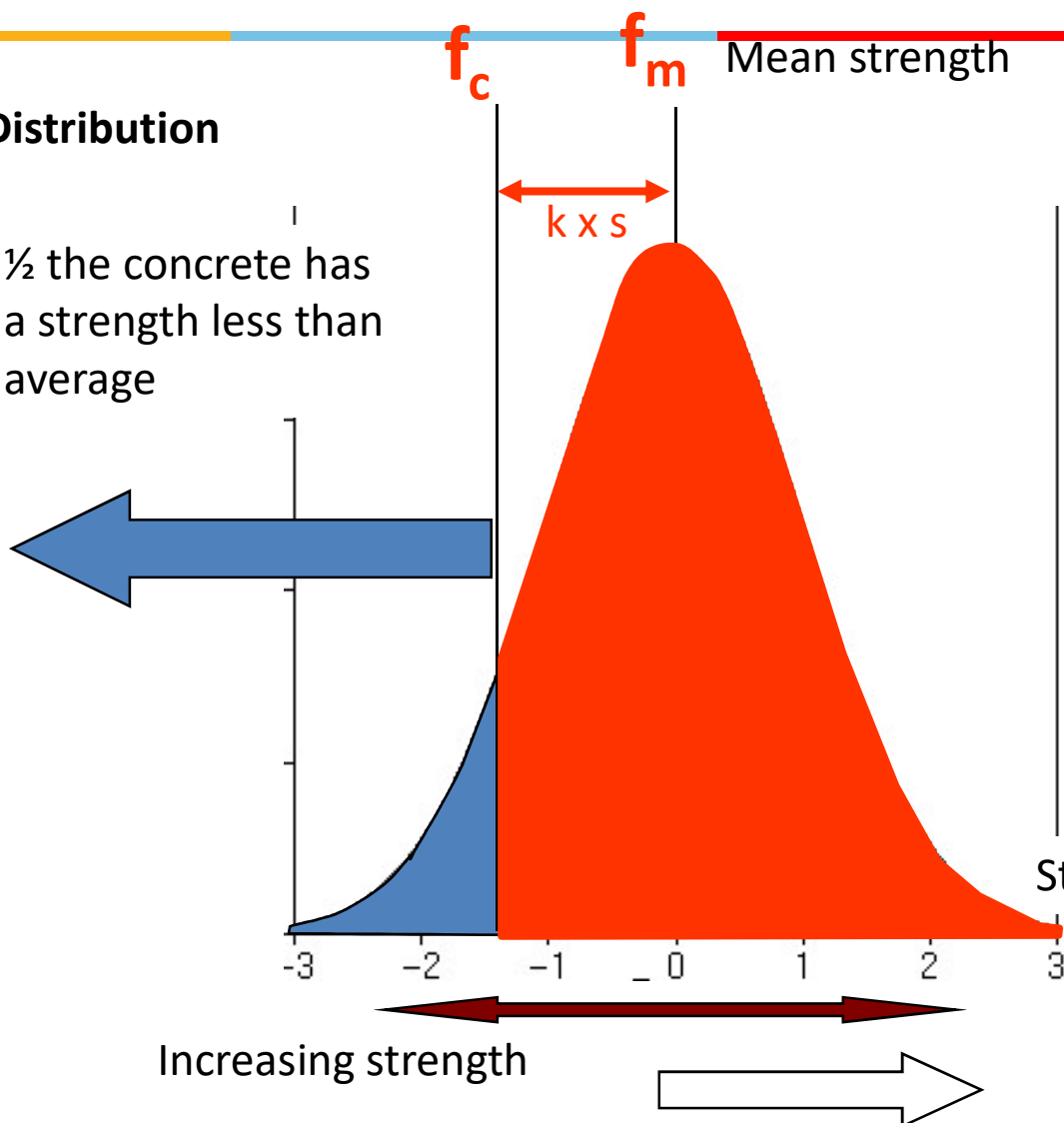
x	$(x - \bar{X})^2$
23.8	0.67
24.3	1.73
20.7	5.21
26.2	10.35
24.1	1.25
23.4	0.17
26.8	14.57
22.7	0.08
19.4	12.84
23.4	0.17
21.5	2.20
19.5	12.13
sum	275.8 61.38



Distribution of Strength

Normal Distribution

½ the concrete has a strength less than average



Mean strength

$$f_m = f_c + k \times s$$

f_m : target mean strength for mix design

f_c : specified characteristic strength

k : probability factor (defective level)

s : standard deviation

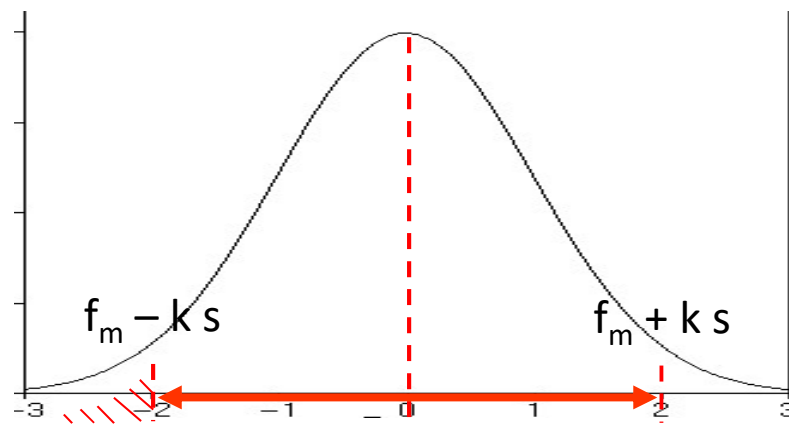
Standard deviations

Increasing strength

Probability Factor as a Function of Defective Level

Table 1: Probability factor as a function of defect level

Probability factor, k	Probability of strength in the range $f_m \pm k s$, %	Probability of strength below $f_m - k s$, %
1.00	68.2	15.9 (1 in 6)
1.64	90.0	5.0 (1 in 20)
1.96	95.0	2.5 (1 in 40)
2.33	98.0	1.0 (1 in 100)
3.00	99.6	0.15 (1 in 700)





Step 1: Determining target strength for mix proportioning

$$f_m = f_c + k \times s$$

f_m = target average compressive strength at 28 days

f_c = characteristic compressive strength at 28 days

s = standard deviation (as specified or from Table 1)

k = 1.65 for 95% probability of strength not less than f_c



Step 2: Selection of w/c ratio

- From table 5 of IS 456, maximum w/c ratio is found out.

Table 5 Minimum Cement Content, Maximum Water-Cement Ratio and Minimum Grade of Concrete for Different Exposures with Normal Weight Aggregates of 20 mm Nominal Maximum Size

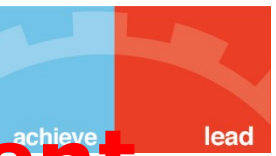
(Clauses 6.1.2, 8.2.4.1 and 9.1.2)

Sl No.	Exposure	Plain Concrete			Reinforced Concrete		
		Minimum Cement Content kg/m ³	Maximum Free Water-Cement Ratio	Minimum Grade of Concrete	Minimum Cement Content kg/m ³	Maximum Free Water-Cement Ratio	Minimum Grade of Concrete
1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Mild	220	0.60	–	300	0.55	M 20
iii)	Moderate	240	0.60	M 15	300	0.50	M 25
iii)	Severe	250	0.50	M 20	320	0.45	M 30
iv)	Very severe	260	0.45	M 20	340	0.45	M 35
v)	Extreme	280	0.40	M 25	360	0.40	M 40

NOTES

1 Cement content prescribed in this table is irrespective of the grades of cement and it is inclusive of additions mentioned in 5.2. The additions such as fly ash or ground granulated blast furnace slag may be taken into account in the concrete composition with respect to the cement content and water-cement ratio if the suitability is established and as long as the maximum amounts taken into account do not exceed the limit of pozzolona and slag specified in IS 1489 (Part 1) and IS 455 respectively.

2 Minimum grade for plain concrete under mild exposure condition is not specified.



Step 3: Selection of water content

- ✓ From Table 4, maximum water content is found out. This value is for angular aggregates and 25-50 mm slump range.
- ✓ Water content is adjusted based on type of aggregate and slump requirement (refer to IS 10262)
- ✓ Water content is further adjusted if water-reducers or superplasticizers are used

Condition	Correction
Sub-Angular Aggregates	- 10 Kg
Gravel + Crushed Particles	- 20 Kg
Rounded Gravel	- 25 Kg
For every slump increase of 25 mm	+ 3 %
Use of Water Reducing Admixture	- 5 to 10 %
Use of Superplasticizing Admixtures	-20 % and above



Step 4: Selecting Cement content

- ✓ Cement and SCM content per unit volume of concrete is calculated from the free w/c ratio and quantity of water calculated in steps 2 and 3
- ✓ Above value of cement content is checked against any specified min. or max. cement content (for durability)

Step 5: Determining proportion of volume of coarse and fine aggregates



- ✓ Table 5 is used to estimate volume proportion of coarse aggregates. This Table is for $w/c = 0.5$
- ✓ Should be suitably adjusted as shown in solved example for a different w/c ratio and if pumpable concrete is used

Step 6: Mix calculations



Find out volume of following ingredients per cubic meter of concrete

Ingredient	Volume (per m ³ of concrete)
Cement	
Water	
Chemical admixture	
All in aggregate (coarse + fine)	

Find out mass of coarse and fine aggregates

Ingredient	Mass (per m ³ of concrete)
Coarse aggregate	
Fine aggregate	